

ORI Projects Meetup

Haifuraiya Channelizer and Power Detection

2 June 2026

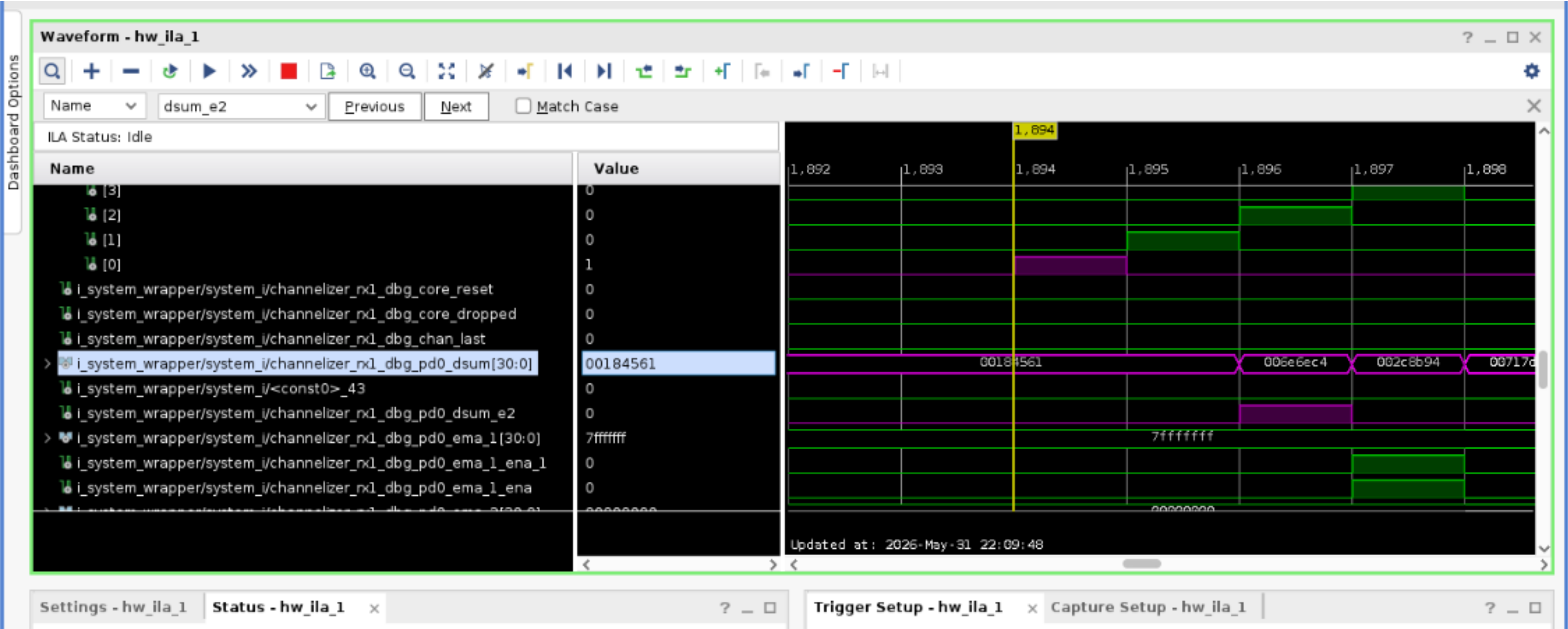
What's Been Done

Work over the past week

- Timing failures fixed (remaining timing failure is not us, it's ADI)
- Exponential Moving Average design bug for larger widths fixed
- Sample rate mismatch rigorously defined and addressed with 2:1 decimator and band equalizer
- Python models and test benches expanded with 2:1 decimator and band equalizer
- End to end test bench updated for 2:1 decimator and band equalizer
- Visualization fully realized and renamed (Takadono to Bouro)
- Discovery: FFT needs to be pipelined, it's dropping frames already

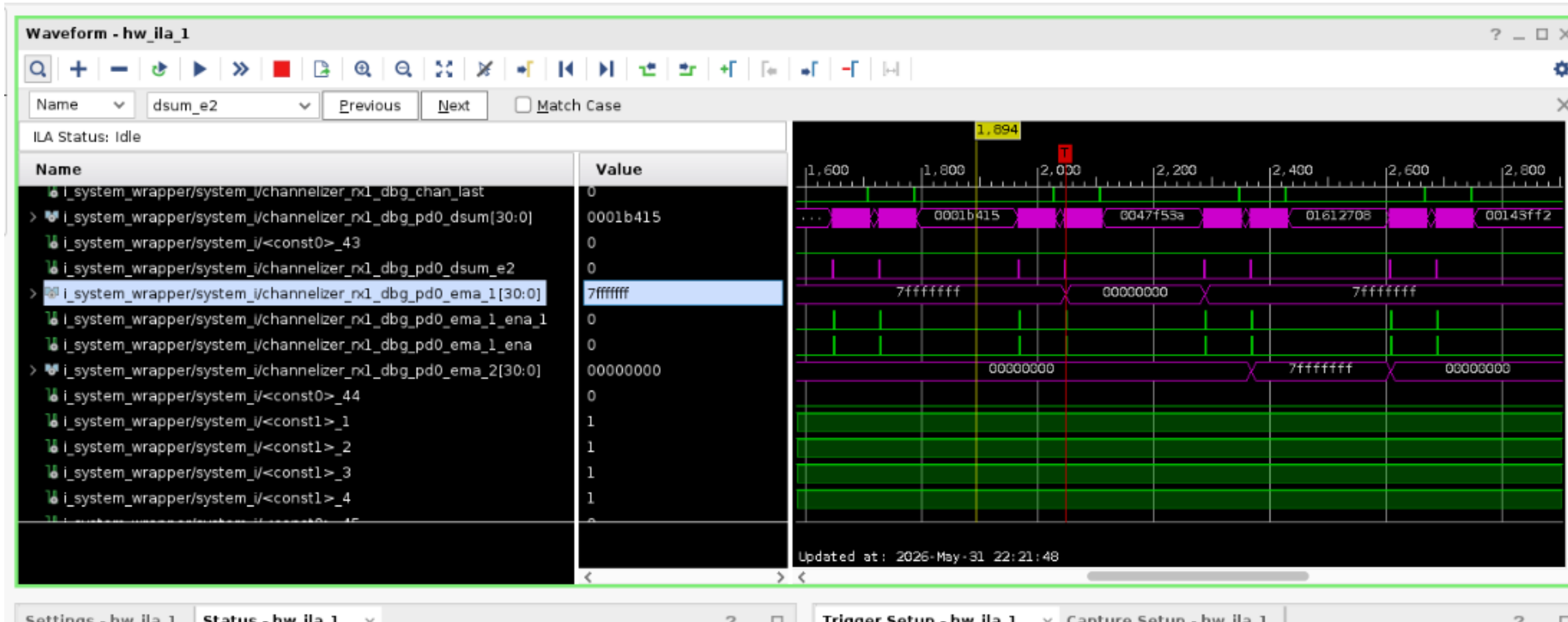
Bugs, But Where

Power Detector Either Saturated or Zero



Backing Out, it's Consistent

Digging in, why did it do this?



Root Cause

Fix, and it's backwards compatible with pluto_msk

The combinational sum assignment summed two shift_left signed terms without saturation, which could wrap past $\pm 2^{(\text{PROD_W}-1)}$ when sustained high-amplitude inputs drove the accumulator past the signed PROD_W-bit range. Once wrapped, the cascade output flipped negative and the EMA settled into a stable wrong-sign equilibrium with no recovery.

Symptom in our use case (polyphase channelizer power detector, AMSAT-UK FunCube+ MDT-SIC): channel-0 EMA-2 sum oscillated between near $+2^{42}$ and near -2^{42} every $\sim 280\mu\text{s}$ under sustained DC input, ending tests in a negative-pinned state that contaminated subsequent measurements.

Fix: compute the sum in EXTRA_W=4 wider intermediate (sum_wide), then clamp to $\pm (2^{(\text{PROD_W}-1)} - 1)$ before assigning to sum. Preserves bit-exact behavior for in-range values; clamps gracefully at the limits.

Verified: full 1ms simulation across 64 EMA cascades shows MSB stays 0 throughout, no wrap events, all 9 testbench assertions pass.

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Why didn't this show up before?

Simulation: silently fixes it

Vivado synth: silently avoids it

GHDL: flagged it as an error

“Deployed” style test bench confirmed

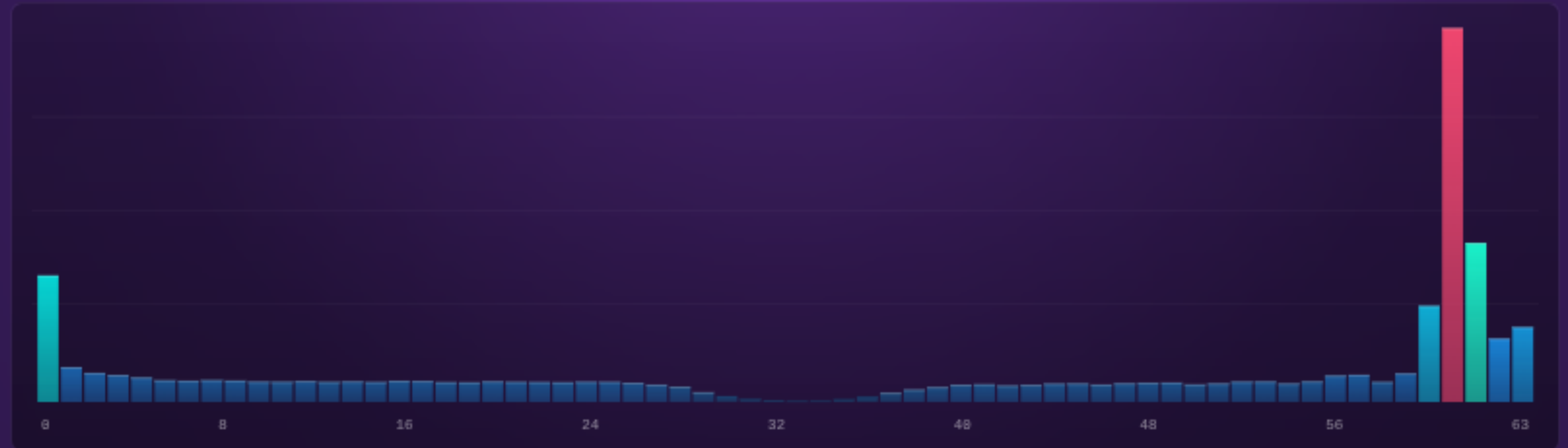
Sample Rate Mismatch

Profile vs. HDL

- VHDL designed for 10 Msps complex sampling for 10 MHz RF bandwidth
- TES profile rejected 10 Msps rates but allowed 20 Msps so we went with it
- After the EMA bug was cleared, we proved we could see powers in the channels. This was a big deal, but the sample rate mismatch meant the design was fundamentally broken.
- The fix was a decimating filter to match what TES would produce with what the polyphase filter bank design was meant to do, at the correct rates for multi-rate work.

Lot going on here

● POWER SPECTRUM · 64 CHANNELS



last update 12:22:21 PM · peak channel 60 · peak value 73,978,653

log scale

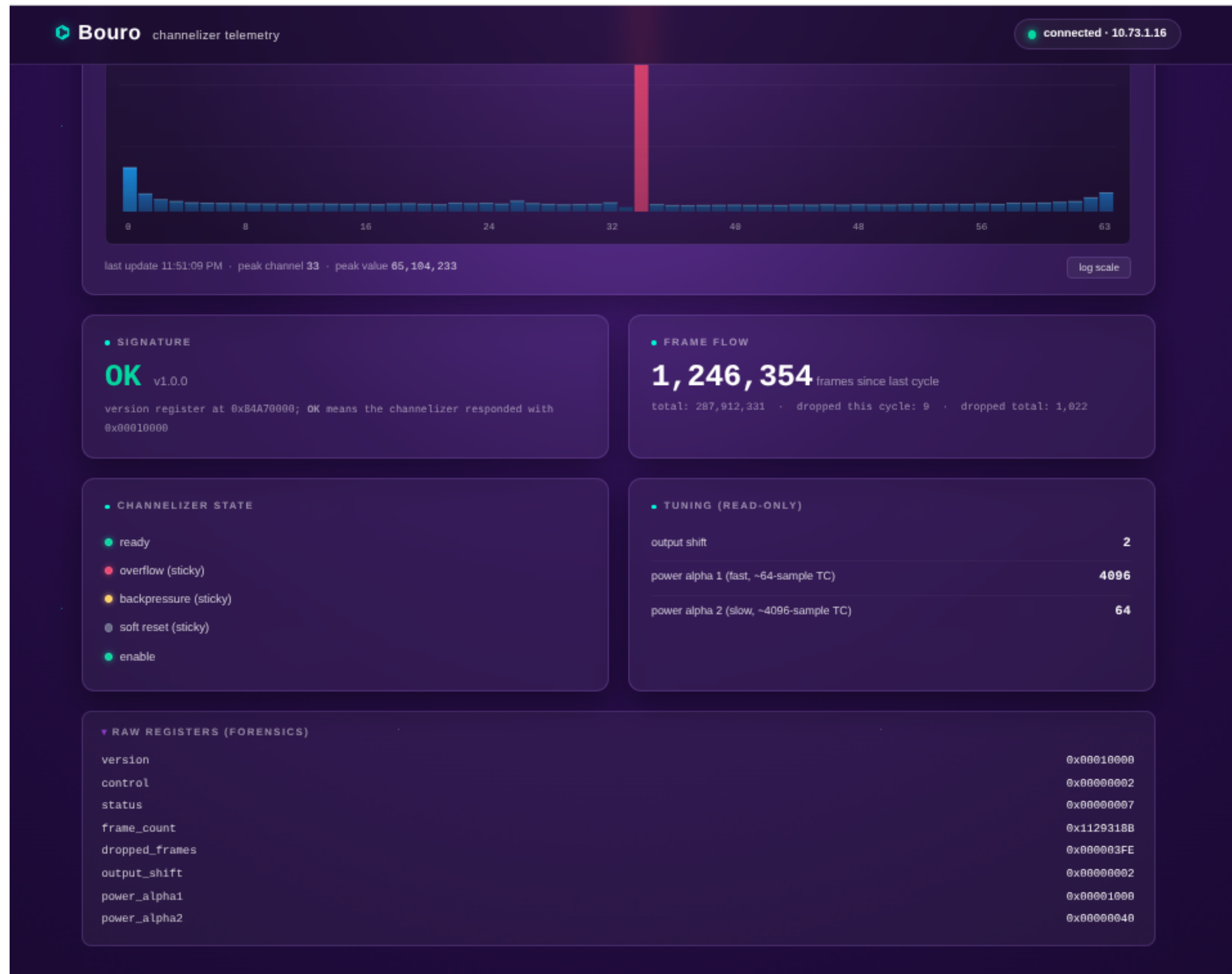
DC spike

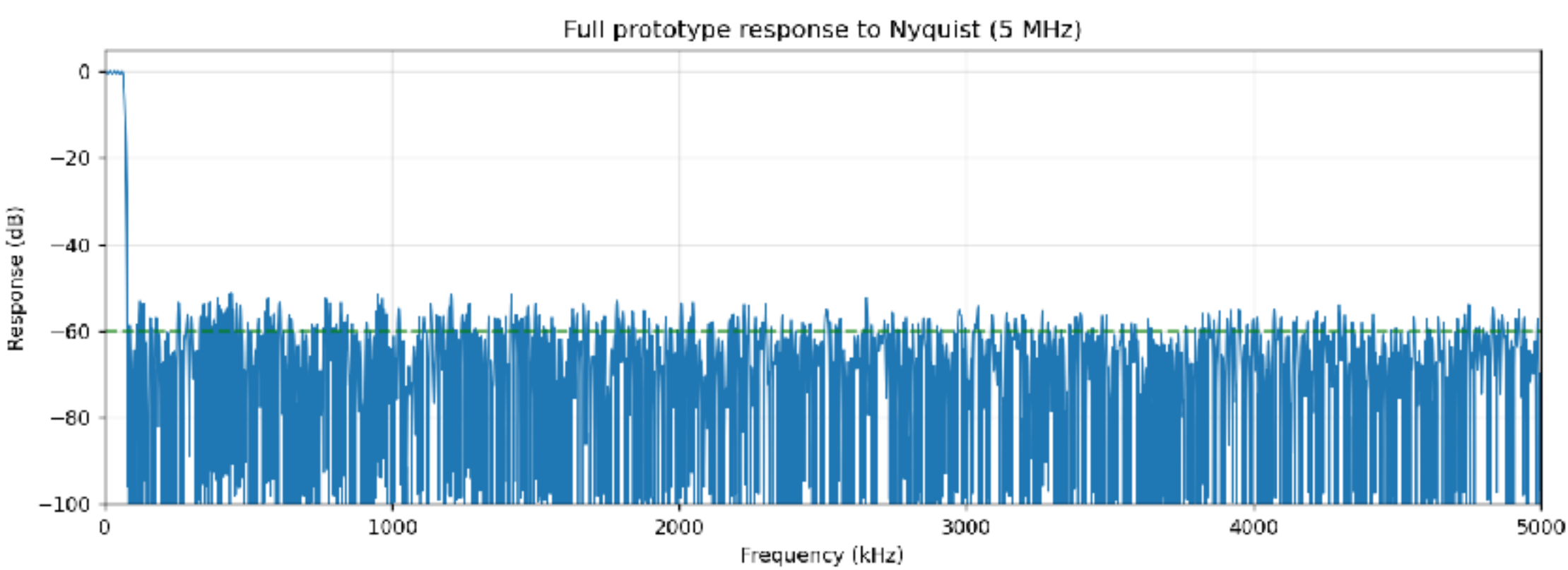
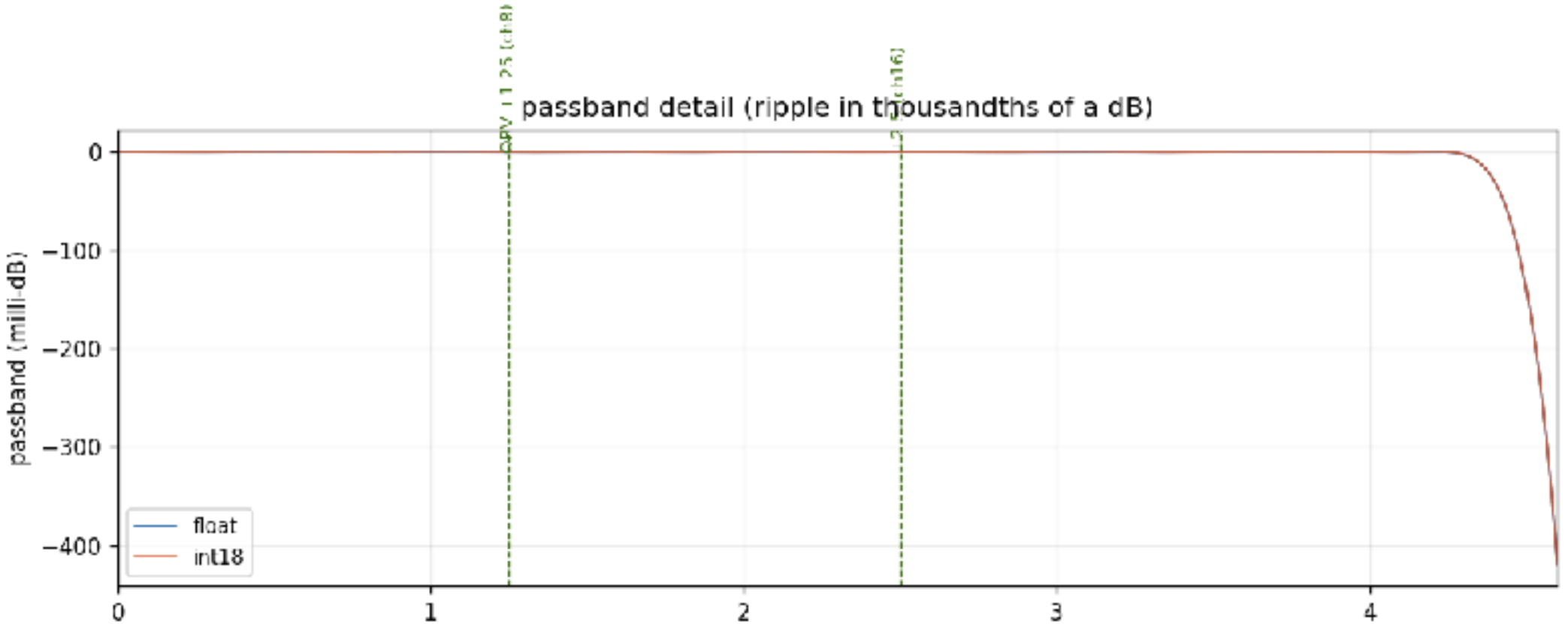
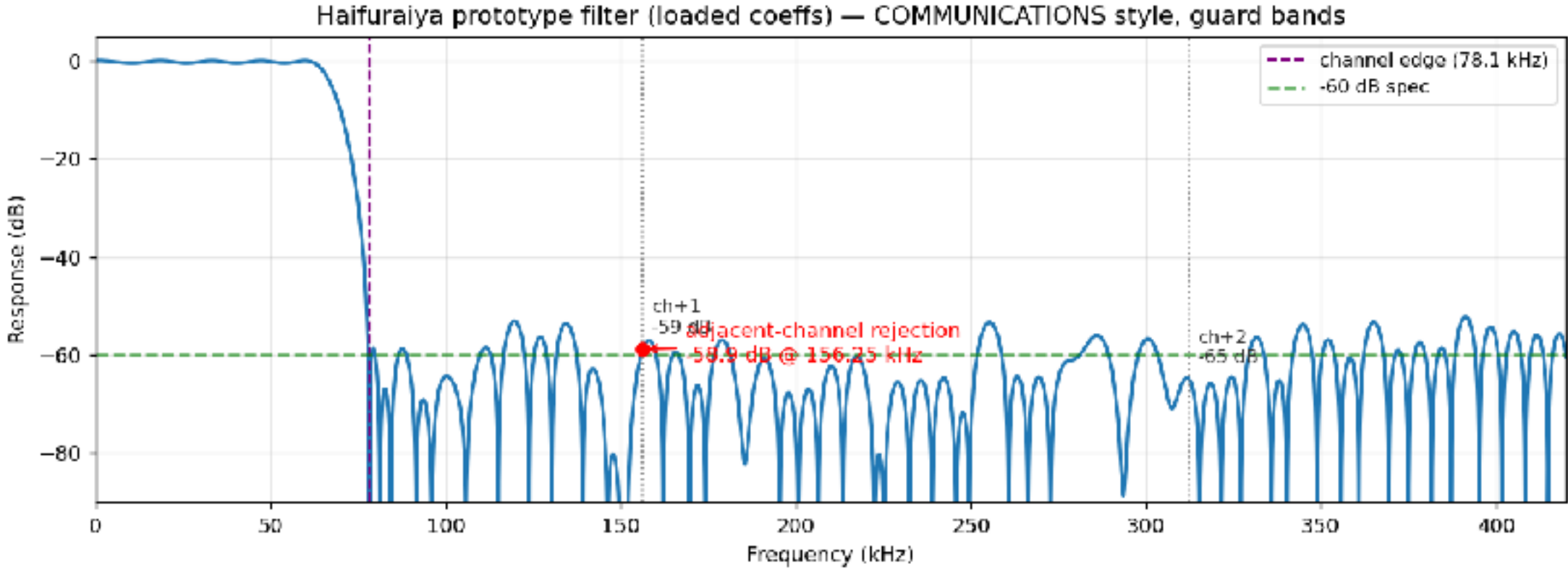
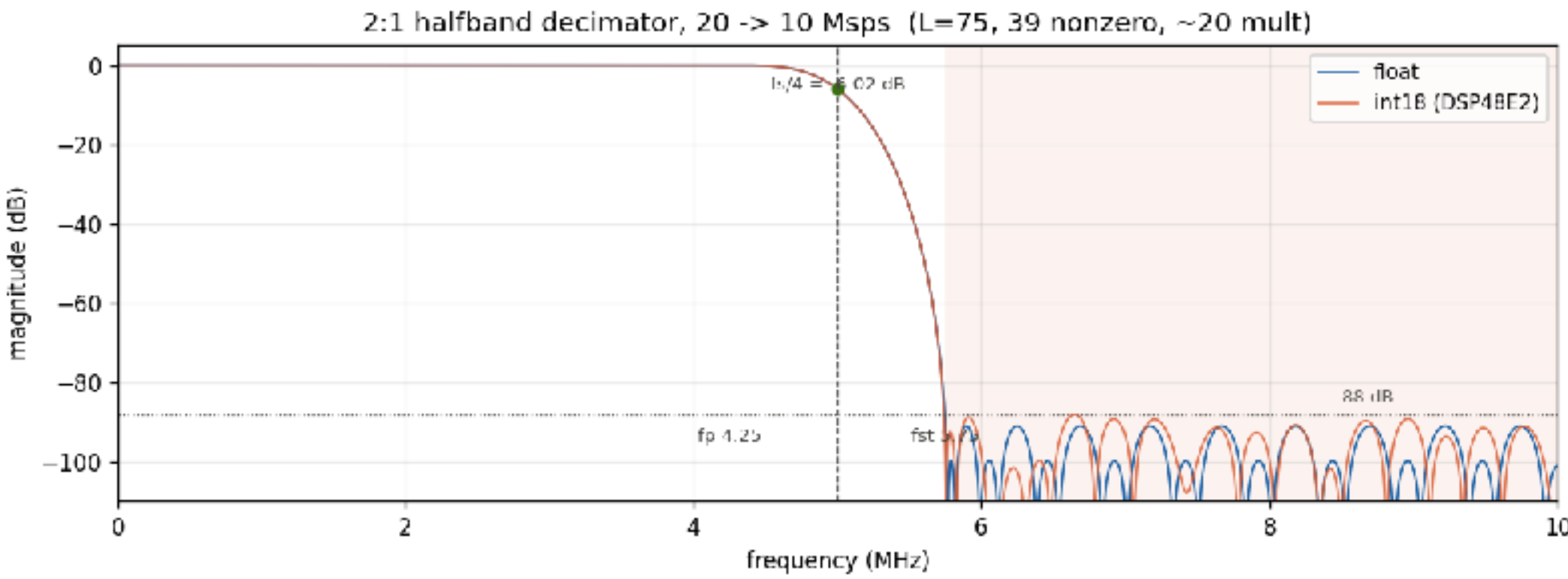
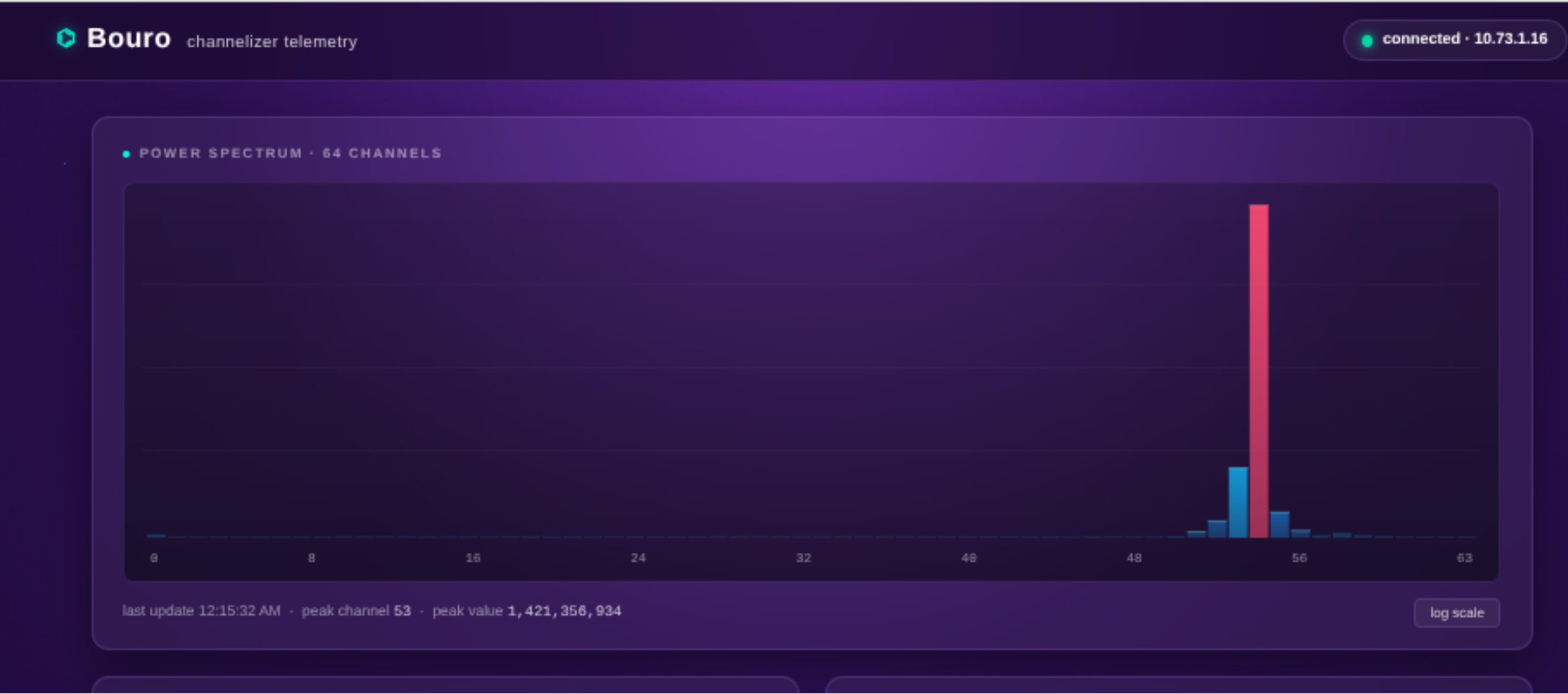
Band edge

Signal!

Possibly with EMA smearing

After the decimating filter and equalization





Next?

Plenty to do, but we are at a significant point

- Real signal energy seen in the channelizer (full characterization next)
- Serious bug solved in EMA submodule
- FFT pipelining dramatically increases resiliency and usefulness of the design
- Demod/decode on the processing side *and* programmable logic side (dual exploration)
- Continuing timing failure in ADI's SERDES reset signal at max clock rate